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Q1. Overfitting happens when a model learns the training examples too well, including random noise, so it shows very low training error but high error on validation/test data.
1) L1/L2 regularization - adds a penalty for large weights so the model becomes simpler.
2) Data augmentation - creating more varied training examples (flips, crops) so the model can't just memorise them.

Q2. Each token gets Q, K and V vectors by multiplying its embedding with weight matrices that are learned in training. Score = Q.K for every pair of tokens, then softmax converts the scores into weights that sum to one, and the output is the weighted sum of the V vectors. We scale by sqrt(dk) to keep the numbers small.

Q3. P = 40/(40+10) = 0.8
R = 40/(40+20) = 0.67
F1 = 2*0.8*0.67/(0.8+0.67) = 0.73
Accuracy = (40+30)/100 = 0.7

(continued)

Q4. GAN: a generator and a discriminator play a minimax game. Diffusion: the model learns to denoise images that were corrupted with Gaussian noise over many timesteps.
GAN training is unstable and can have mode collapse; diffusion training is stable and the samples are more diverse.
GANs are fast (a single forward pass), diffusion is slow (hundreds of denoising steps).

Q5. RAG: chunk the documents, embed each chunk with an embedding model and store the vectors in a vector DB like FAISS. At query time embed the question, retrieve the top-k similar chunks and put them in the prompt for the LLM.
Failure: chunks too large or too small, so the right information is split or lost -> tune the chunk size and use overlap between chunks.
Another issue is cost.